



infinlTy Online Camp JEE (Adv) 2024

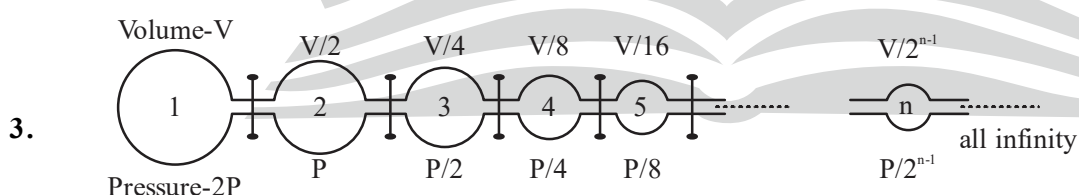
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SUBJECT: CHEMISTRY

DATE: 09-04-2024

- The van der Waal's constant 'b' for N_2 and H_2 has the values $0.04 \text{ lit mol}^{-1}$ and $0.025 \text{ lit mol}^{-1}$. The density of solid N_2 is 1 g cm^{-3} . Assuming the molecules in the solids to be close packed with the same percentage void, the density of solid H_2 would be (in g cm^{-3})
 (A) 0.114 (B) 0.682
 (C) 1.466 (D) 0.071
- If the number of molecules of SO_2 (molecular weight = 64) effusing through an orifice of unit area of cross-section in unit time at 0°C and 1 atm pressure is n. The number of He molecules (atomic weight = 4) effusing under similar conditions at 273°C and 0.25 atm is
 (A) $\frac{n}{\sqrt{2}}$ (B) $n\sqrt{2}$
 (C) $2n$ (D) $\frac{n}{2}$



Infinite number of flasks are connected to one another as shown above. The volume and pressure in each flask vary as shown. The stopcocks are initially closed. The common pressure, when all the stopcocks are opened, is: (Assume constant temperature)

- (A) P (B) $\frac{1}{2} P$ (C) $\frac{P}{4}$ (D) $\frac{4}{3} P$
- At a certain temperature for which $RT = 25 \text{ lit. atm. mol}^{-1}$, the density of a gas, in gm lit^{-1} , is $d = 2.00 P + 0.020 P^2$, where P is the pressure in atmosphere. The molecular weight of the gas in gm mol^{-1} is
 (A) 25 (B) 50
 (C) 75 (D) 100

5. The densities of three gases O_2 , SO_2 and H_2 are measured at 400 K and 1 atm. They are 0.8 g/L , $\frac{32}{15} \text{ g/L}$ and $\frac{1}{25} \text{ g/L}$ respectively. Which of the following statements is/are correct for these gases? (Use $R = \frac{1}{12} \text{ L atm/mol-K}$).
- (A) SO_2 is showing negative deviation from ideal behaviour.
- (B) O_2 is more compressible than ideal gas under these conditions.
- (C) H_2 is behaving ideally.
- (D) Boyle's temperature of O_2 is less than 400 K.

Paragraph Ques. no. 6 to 8

The system shown in the figure is in equilibrium, where A and B are isomeric liquids and form an ideal solution at T_K . standard vapour pressure of A and B are P_A° and P_B° respectively at T_K . We collect the vapour of A and B in two container of volume V, first container is maintained at $2T_K$ and second container is maintained at $\frac{3T}{2} K$. At the temperature greater than T_K , both A and B exist in only gaseous form. Total vapour pressure of the system at T_K is given as

$$P = P_A^\circ X_A + P_B^\circ X_B$$

where X_A and X_B are the mole fraction of A and B in a liquid mixture.

In container (I)

We assume that collected gases behave ideally at $2T_K$ and there can take place isomerisation reaction in which A converted into B by first order kinetics

Reaction is given as



In container (II)

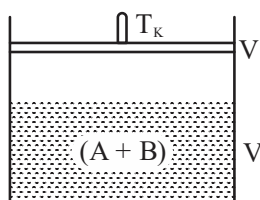
At the given temperature $\frac{3T}{2}$, A and B are ideal in nature and non mixing in nature. A small pin hole is made into container. We can determine the initial rate of effusion of both gases in vacuum by the expression

$$r = \frac{kp}{\sqrt{m}}$$

Where, P = pressure difference between system and surrounding

K = positive constant

M = Molecular weight of the gas.



6. If partial vapour pressure of A is twice that of partial vapour pressure of B and total V.P. is equal to 2 at $T = 50\text{ K}$ and $V = 8.21\text{ lit}$, then number of mole of A and B in vapour phase is :
- (A) $\frac{8}{3}, \frac{4}{3}$ (B) 3,1 (C) 2,2 (D) 1,3
7. Vapour is collected and passed into a container of volume 8.21 lit, maintained at 100 K and after 5 min number of mole of B = $\frac{8}{3}$. Then calculate pressure developed into the container after two half lives.
- (A) 2 atm (B) 4 atm (C) 1 atm (D) 0.5 atm
8. If vapour is collected in container of volume 8.21 lit maintained at 75 K then ratio of initial rate of effusion of Gas A and B is given as:
- (A) 2 : 1 (B) 1 : 2 (C) 4 : 1 (D) 1 : 4

ANSWER KEY

Ques.	1	2	3	4	5	6	7	8			
Ans.	A	A	D	B	AD	A	A	A			